Food preservation myth busters — Volunteer key

FOOD SAFETY

06-FoodSafety-001 **TRUE** **MEDIUM RISK**

Myth: It is safer to refrigerate vacuum-packaged low-acid foods at home rather than freeze or can them.

Bust: Vacuum packaging removes oxygen, which slows mold and bacterial spoilage, but low-acid perishable foods still need refrigeration to stay safe. Without refrigeration, dangerous bacteria can grow even in vacuum packaging. Freezing or canning are necessary for longer term storage.

Why believed: Many people trust vacuum packaging alone to preserve food safely and think refrigeration may not always be needed.

The science: Vacuum packaging alone does not kill bacteria but reduces oxygen, limiting growth of aerobic bacteria. However, anaerobic pathogens like Clostridium botulinum can grow if food is not refrigerated. Thus, low-acid vacuum-packed foods must be kept cold, ideally at 38-40°F, to prevent pathogen growth .

Source: SP 50-1000 Turkey basics

06-FoodSafety-002 **TRUE** **MEDIUM RISK**

Myth: Using pasteurized eggs in recipes with raw or lightly cooked eggs safely prevents foodborne illness.

Bust: Recipes like homemade mayonnaise or eggnog containing raw or lightly cooked eggs can be a risk for Salmonella infection. Using pasteurized eggs or egg products eliminates this risk and keeps your food safe.

Why believed: Many people hesitate to change traditional recipes, fearing taste or quality loss, or overlook the risk from raw eggs.

The science: Pasteurization heats eggs just enough to kill Salmonella without cooking the egg, making them safe for uncooked or lightly cooked uses. Raw eggs may harbor Salmonella even if shells look clean.

Source: PNW 250 You can prevent food borne illness

06-FoodSafety-003 **FALSE** **HIGH RISK**

Myth: Freezing home-canned foods preserves them safely for long-term storage without any risk.

Bust: Freezing canned foods is not a safe preservation method. Home-canned foods must be processed and stored properly at room temperature. Freezing can cause jars to crack or lids to loosen, which risks contamination. Additionally, freezing does not kill bacteria like botulism spores that can grow in canned low-acid foods.

Why believed: People often think freezing everything halts all bacteria because it slows bacterial growth. Also, reusing home-canned foods for freezing seems convenient and logical.

The science: Pressure canning food kills Clostridium botulinum spores by processing at 240°F. Freezing does prevent bacteria from growing temporarily but does not destroy spores. Food safety guidelines warn against freezing canned low-acid foods because of jar breakage and incomplete bacterial kill. Proper canning methods followed by storage at room temperature are essential to prevent botulism .

Source: PNW 361 Canning meat poultry game

06-FoodSafety-004 **FALSE** **HIGH RISK**

Myth: You don't need to worry about flour as a food safety risk since it's dry and uncooked.

Bust: Flour can contain harmful bacteria like E. coli, and because it is raw and dry, these bacteria are not killed until the flour is cooked. Eating raw dough or batter made with uncooked flour can cause foodborne illness. It's important to avoid tasting raw dough or batter and to cook flour-containing foods thoroughly.

Why believed: People see dry flour as stable and safe and associate cooking safety only with meats or fresh produce, leading to underestimating flour's risks.

The science: Raw flour is a raw agricultural product and can harbor pathogens. Outbreaks linked to raw flour have been confirmed. Cooking or baking foods to proper temperatures kills bacteria. Food safety guidelines warn against consuming raw dough or batter containing untreated flour to prevent illness .

Source: PNW 717 Dangers lurking in your flour

06-FoodSafety-005 **FALSE** **HIGH RISK**

Myth: You can safely pasteurize raw milk at home by heating it briefly to a lower temperature than store-bought pasteurization.

Bust: Home pasteurization attempts often use too low a temperature or too short a time, failing to kill all harmful bacteria. Proper pasteurization requires heating milk to at least 161°F for 15 seconds or 145°F for 30 minutes. Unsafe home pasteurization may give a false sense of security and put people at risk for serious illness.

Why believed: Some people think they can replicate commercial pasteurization with home heating methods to enjoy raw milk

benefits while being safe.
The science: Official pasteurization follows specific time and temperature combinations validated to destroy pathogens. Heating milk incorrectly leaves surviving bacteria including Listeria, Salmonella, and E. coli. Home pasteurization methods without accurate controls are unsafe and not recommended .
Source: SP 50-1000 Turkey basics

CANNING FRUITS

06-Fruits-001 **TRUE** **LOW RISK**

Myth: The hot pack method is better than the raw pack method for canning fruits because it helps preserve color and shortens processing time.

Bust: Hot pack involves briefly boiling fruits before packing in jars, which helps retain color, flavor, texture, and reduces floating fruit. It also usually shortens processing time compared to raw pack. Though some prefer raw pack for firmer texture, hot pack has distinct advantages.
Why believed: Some cooks doubt heating fruit first, thinking it reduces nutrient quality or texture. Traditional methods often favored raw packing for fruit firmness.
The science: The hot pack method heats fruit, initiates enzyme inactivation, and reduces trapped air. This improves color retention and packing efficiency, as noted in OSU guidelines. Processing times are often shorter because fruit is already partially cooked.
Source: PNW 199 Canning fruits

06-Fruits-002 **TRUE** **MEDIUM RISK**

Myth: Adding extra bottled lemon juice or citric acid to fruits like figs before canning is unnecessary and can make the fruit too sour.

Bust: Adding bottled lemon juice or citric acid is necessary for safety when canning low-acid fruits such as figs and Asian pears. Without acidification, the risk of botulism increases. Using the recommended amounts ensures safety without causing excessive sourness.
Why believed: Many find the added acid makes the fruit taste too tart, so they skip it, influenced by tradition or taste preferences.
The science: Low-acid fruits need acid added to lower pH to safe levels for boiling water canning. Bottled lemon juice or citric acid is standardized for acidity unlike fresh lemons. Without this, spores of Clostridium botulinum may survive and produce toxin.
Source: PNW 172 Canning vegetables

06-Fruits-003 **FALSE** **HIGH RISK**

Myth: Skipping adding lemon juice or other acidifiers when canning low-acid fruits like figs or Asian pears has no impact on safety.

Bust: Acidifying low-acid fruits such as figs and Asian pears with lemon juice before canning is important to make the product safe. Without this step, harmful bacteria may survive in the low-acid environment. Adding the recommended amount of bottled lemon juice helps ensure the acidity is high enough for safe water bath canning.
Why believed: Some people rely on old family recipes or assume that fruits are naturally acidic enough to be safe without added acid. Traditional recipes may omit acidifiers because taste or texture was prioritized over safety in the past.
The science: Low-acid fruits have pH above 4.6, which allows pathogens like Clostridium botulinum to survive unless acidified. The USDA and OSU guidelines recommend adding specific amounts of bottled lemon juice or citric acid to lower pH to safe levels for water bath canning. Without this step, high risk of botulism exists.
Source: SP 50-1003 Laws of salsa with recipes

06-Fruits-004 **FALSE** **HIGH RISK**

Myth: You can safely reuse canning lids as long as they look new and undamaged.

Bust: Reusable metal bands can be used multiple times, but flat canning lids should never be reused. Each lid has a sealing compound designed for one use only. Reusing lids risks poor seals, leading to spoilage or dangerous microbial growth inside the jar.
Why believed: Some people want to save money or avoid waste by reusing lids. This might come from reusing other food packaging or from misunderstanding instructions from earlier home canning practices.
The science: Manufacturers design lids with single-use sealing compound. Reuse compromises the seal and increases risk of contamination. OSU Extension and USDA recommend always using new lids for safe home canning.
Source: PNW 199 Canning fruits

06-Fruits-005 FALSE HIGH RISK

Myth: Filling jars completely to the top with fruit and liquid when canning fruits ensures a better seal and safer product.

Bust: It is important to leave the recommended headspace (usually 1/2 inch) between the fruit or liquid and the jar rim. Overfilling can cause food to bubble out during processing, interfering with the seal and potentially leading to spoilage or unsafe canned goods.

Why believed: People may think max filling is better to store more food or prevent air exposure. This belief may come from misunderstanding headspace importance from older cookbooks or visual intuition.

The science: Headspace allows air to escape during processing and creates a vacuum seal upon cooling. Too little headspace causes seal failure; too much may prevent vacuum. OSU guidelines emphasize the specific headspace for safe canning success.

Source: PNW 199 Canning fruits

CANNING TOMATOES & SALSAS

06-Tomatoes-001 TRUE MEDIUM RISK

Myth: Adding bottled lemon juice or vinegar to canned tomatoes is necessary to ensure their safety.

Bust: Tomatoes have borderline acidity, with pH near 4.6, so adding bottled lemon juice or vinegar provides consistent acidity to prevent harmful bacteria growth. This acidification ensures tomatoes can be safely processed in a boiling water canner. Without added acid, tomatoes may not be safe to can.

Why believed: This practice is common in many canning guides and home canning traditions, where natural tomato acidity varies and bottled acid is a reliable way to ensure safety.

The science: Botulism bacteria cannot grow in sufficiently acidic environments (pH below 4.6). Because tomato acidity varies, adding bottled lemon juice or vinegar standardizes the acidity level to safely inactivate pathogens during boiling water bath processing.

Source: PNW 300 Canning tomatoes and tomato products

06-Tomatoes-002 TRUE LOW RISK

Myth: It's safe to can green tomatoes in salsa recipes designed for red tomatoes without changing the amount of acid or processing time.

Bust: Green tomatoes can be used in place of red tomatoes in tested salsa recipes as long as you do not alter the acid amounts or processing times. Some research-tested recipes even call for green tomatoes, maintaining the safety of the finished product by keeping correct acidity levels intact.

Why believed: People often think green tomatoes are less acidic or unsafe because they are unripe, leading to doubts about including them in salsa.

The science: Tested recipes consider the acidity of green tomatoes and provide appropriate vinegar or lemon juice to maintain safe pH levels to prevent botulism during boiling water canning.

Source: PNW 172 Canning vegetables

06-Tomatoes-003 FALSE HIGH RISK

Myth: You can safely can your own original salsa recipe by adjusting ingredients to taste without following a tested recipe.

Bust: Salsas are acidified foods combining low-acid and acid ingredients. Only tested recipes have the proper acidity to prevent dangerous bacteria growth. Creating your own recipe risks unsafe acidity levels that can cause botulism. Always use research-tested recipes for canning salsa or freeze your original recipe instead.

Why believed: Many people trust family recipes or like to customize flavors, believing that traditional methods or personal tweaks are safe because it 'cooked down' or 'boiled enough.'

The science: Salsa acidity level is critical for safety due to Clostridium botulinum risk. Non-tested recipes may lack sufficient acid, requiring lab testing to verify safety processing times and methods. Without this, botulinum toxin-producing bacteria can survive and multiply.

Source: PNW 172 Canning vegetables

06-Tomatoes-004 FALSE HIGH RISK

Myth: Boiling salsa before filling jars reduces processing time or eliminates the need for proper canning processing.

Bust: Boiling salsa before jar filling is part of some recipe steps but does not replace the required boiling water bath or pressure canning process. Proper processing time in a water bath canner is crucial to eliminate bacteria and ensure

safety. Skipping or shortening processing times puts you at risk of botulism.
Why believed: People may think that making salsa hot or boiling it first 'kills everything,' so extra processing is unnecessary, especially in homemade batches.
The science: Pathogenic bacteria and spores, including botulinum spores, require the precise processing time and temperature during canning to be destroyed. Boiling salsa on the stove does not maintain these conditions in sealed jars, so it cannot replace official processing.
Source: PNW 395 Salsa recipes for canning

06-Tomatoes-005 **FALSE** **MEDIUM RISK**

Myth: Adding extra vinegar to a tested salsa recipe makes it safer and can't harm the final product's safety.
Bust: While it might seem adding more vinegar increases safety, altering acid levels beyond what tested recipes specify can disrupt the product's texture, flavor, and pH balance. It can potentially lead to unsafe preservation results or inferior quality. Always follow the exact acid amounts in tested recipes.
Why believed: People intuitively think 'more acid equals safer food' and may add extra vinegar for tangy flavor or perceived extra safety.
The science: Canning recipes are carefully tested for acidity and processing times to guarantee safety. Deviating from these acid amounts can interfere with heat penetration or cause poor quality, making the product unsafe despite higher acid.
Source: PNW 172 Canning vegetables

LOW ACID FOODS

06-LowAcid-001 **TRUE** **MEDIUM RISK**

Myth: Boiling home-canned low acid foods before eating is an extra safety step that helps destroy any botulism toxin present.
Bust: Boiling low acid home-canned foods for 10 minutes before eating is an important safety measure to destroy any botulism toxin that might be present. This extra step is recommended because botulism toxin can be deadly even if the canned food looks normal.
Why believed: Some doubt boiling is necessary if the food looks, smells, and tastes fine. Boiling is considered extra precaution by some, but skepticism exists because it changes food quality.
The science: Botulism toxin can develop in improperly processed low acid foods and can survive normal cooking temperatures. Boiling for 10 minutes deactivates the toxin and prevents poisoning.
Source: PNW 361 Canning meat poultry game

06-LowAcid-002 **TRUE** **MEDIUM RISK**

Myth: Soaking and cooking dried beans before pressure canning them is necessary to ensure safety and quality.
Bust: Soaking and pre-cooking dried beans makes them safe and improves texture when canned. Skipping this step can lead to poor quality and unsafe results.
Why believed: Traditional canning methods sometimes omit soaking, but modern research shows it's essential for safety and quality.
The science: Dried beans and peas must be soaked and cooked before canning to reduce processing time and ensure heat penetrates to destroy bacteria safely.
Source: PNW 450 Canning smoked fish at home

06-LowAcid-003 **FALSE** **HIGH RISK**

Myth: You can safely can low acid foods like meat and vegetables using a boiling water canner instead of a pressure canner.
Bust: Low acid foods such as meat, poultry, and vegetables must be canned using a pressure canner because boiling water canners do not reach high enough temperatures to kill dangerous bacteria like Clostridium botulinum. Using a boiling water canner can leave these foods unsafe, with a risk of botulism poisoning.
Why believed: Many people think that boiling water canners are sufficient because they work well for high acid foods like fruits and pickles. Some traditional recipes or old practices may not clearly advise pressure canning.
The science: Low acid foods have a pH above 4.6, creating an environment where botulism spores can survive and produce toxin. Only pressure canners reach the necessary 240°F (116°C) to destroy these spores. Boiling water reaches only 212°F (100°C), which is insufficient to kill spores in low acid foods.
Source: PNW 361 Canning meat poultry game

06-LowAcid-004 **FALSE** **HIGH RISK**

Myth: Adding a small amount of vinegar to low acid vegetables when canning can make them safe to process in a boiling water canner.

Bust: Simply adding a little vinegar to low acid vegetables does not guarantee they become acidic enough for safe boiling-water canning. The acidity level must be very precise to prevent botulism risk, so home canners should use tested recipes designed for acidified foods or use pressure canning.

Why believed: People often think vinegar is a universal preservative that can make any food safe for water bath canning because vinegar is commonly used in pickling.

The science: The pH must be 4.6 or below to prevent bacterial growth. If the amount of acid added is not adequate throughout the food, spores may survive. Only tested recipes for acidified foods ensure sufficient acid levels for boiling water canner safety.

Source: PNW 361 Canning meat poultry game

06-LowAcid-005 **FALSE** **HIGH RISK**

Myth: Properly venting a pressure canner for 10 minutes before pressurizing is unnecessary if the canner gets hot quickly.

Bust: Venting the pressure canner for 10 minutes is essential to remove air from the canner chamber. Air trapped inside can prevent the proper temperature from being reached, causing under-processing and unsafe canned food.

Why believed: Many may think that once steam starts, the canner is ready, especially if the process seems fast. It can seem like an unnecessary delay.

The science: Air inside the canner lowers the temperature and prevents the jars from reaching 240°F, necessary to kill bacterial spores. Proper venting ensures the canner fills with pure steam and reaches the needed temperature.

Source: PNW 361 Canning meat poultry game

PICKLING

06-Pickling-001 **TRUE** **MEDIUM RISK**

Myth: Fermented pickles require several weeks of curing at room temperature for safety and flavor development.

Bust: Fermented pickles use good bacteria to produce acid gradually, which preserves the vegetables and adds flavor. This process naturally takes time at room temperature.

Why believed: Traditional fermentation has always involved waiting several weeks; it's a heritage method handed down in families and cultures.

The science: During fermentation, lactic acid bacteria grow and produce acid needed to inhibit harmful microbes. This process takes weeks at room temperature to ensure safety and desired flavor.

Source: PNW 355 Pickling vegetables

06-Pickling-002 **TRUE** **LOW RISK**

Myth: Quick pickles can be safely made with reduced salt if there is enough vinegar in the recipe.

Bust: In quick pickles, salt affects flavor and texture but is not critical for safety if the vinegar concentration is maintained properly.

Why believed: It's common to think salt is always necessary for safety, but vinegar is the main acidifying agent in quick pickles.

The science: Quick pickles rely on vinegar acidity to inhibit pathogens. Salt enhances flavor and crispness but can be reduced safely if vinegar remains adequate.

Source: NCHFP Pickled eggs

06-Pickling-003 **FALSE** **HIGH RISK**

Myth: Adding extra vinegar to a pickling recipe always makes the pickles safer.

Bust: Simply adding more vinegar does not guarantee safety if the recipe's balance is off. Proper acid level, recipe proportions, and processing time are critical for safety.

Why believed: People often think that since vinegar is acidic and kills bacteria, more vinegar must be safer. This seems logical and is a common kitchen intuition.

The science: Pickling recipes are carefully formulated to reach a safe pH and preserve texture and flavor. Adding extra vinegar can dilute salt concentration or alter other factors, potentially making the product less safe and of poor quality.

Source: PNW 355 Pickling vegetables

06-Pickling-004 FALSE HIGH RISK

Myth: Using a garlic clove or onion slice in a pickled egg jar reduces the need for proper acidity or refrigeration.

Bust: Flavoring ingredients like garlic or onion do not replace the need for correct vinegar concentration or refrigeration for safety.

Why believed: Family recipes often include these flavorful add-ins and sometimes neglect refrigeration, assuming they add preservation effect.

The science: Only the acidity and proper refrigeration prevent microbial growth in pickled eggs; flavorings do not affect safety and can pose contamination risks if not handled correctly.

Source: NCHFP Pickled eggs

06-Pickling-005 FALSE HIGH RISK

Myth: Salt can be omitted from fermented pickle recipes without affecting food safety.

Bust: Salt is essential in fermented pickles to inhibit spoilage organisms and allow beneficial bacteria to thrive. Omitting salt can lead to unsafe products.

Why believed: Some people reduce salt for health reasons or believe salt is only for flavor, thinking the fermentation alone is sufficient.

The science: Salt concentration controls microbial growth during fermentation. Without proper salt levels, harmful bacteria can grow and spoil or contaminate the product.

Source: NCHFP Pickled eggs

JAMS & JELLIES

06-Jams-001 TRUE LOW RISK

Myth: Adding bottled lemon juice to a jam recipe is necessary to ensure safe acidity and proper setting when using low-acid fruits.

Bust: Adding bottled lemon juice is recommended because it has a reliable acidity level that helps preserve safety by lowering pH and improves gel formation with pectin. Using fresh lemon juice or substituting vinegar can be inconsistent and unsafe.

Why believed: Traditionally, recipes call for lemon juice to balance sweetness and acidity, and home canners know acidity is crucial for safety.

The science: Bottled lemon juice is standardized at pH 2.0-2.6, ensuring consistent acid levels. Proper acidification prevents growth of harmful bacteria and allows commercial pectin to gel effectively.

Source: SP 50-632 Making berry syrups at home

06-Jams-002 TRUE LOW RISK

Myth: It's safer to use only pint or smaller jars for jams and jellies because larger jars may not set properly.

Bust: Using half-pint or pint jars helps jams and jellies set properly because larger jars cool more slowly and may stay too warm too long, causing the gel to fail. So, picking the right jar size is indeed important for a quality and safe product.

Why believed: This advice comes from pectin manufacturers and extension publications that recommend smaller jars for best setting results, but sometimes is misunderstood as a safety issue rather than quality.

The science: Cooling rate affects gel formation; slow cooling in larger jars can prevent the gel from setting firmly, resulting in a runnier product. Smaller jars cool faster and set better, contributing to safer, stable jams.

Source: SP 50-632 Making berry syrups at home

06-Jams-003 FALSE HIGH RISK

Myth: Using homemade fruit pectin instead of commercial pectin in jam recipes always produces safe, high-quality jams.

Bust: While homemade pectin can sometimes be used, commercial pectin products are tested to ensure they work with specified sugar and acid levels for safe gel formation. Homemade or untested pectin substitutions risk under-processing or incorrect gel, leading to spoilage.

Why believed: Traditionally, people used natural methods for thickening jams and assume any pectin source is equally effective.

The science: Commercial pectin is standardized for quality and performance; low methoxyl and other special pectins require specific sugar and acid conditions. Incorrect pectin can cause jams not to set or not to be acidic enough, risking bacterial growth.

Source: SP 50-632 Making berry syrups at home

06-Jams-004 **FALSE** **HIGH RISK**

Myth: It is safe to add thickening agents like cornstarch or flour directly to jams before canning to get a thicker product.

Bust: Adding thickeners before canning is not recommended because they can interfere with heat penetration and preservation, potentially causing unsafe jams. Instead, thicken jams by cooking or thickening after opening.

Why believed: People want to achieve thicker texture quickly and believe adding thickeners is an easy way to improve jam consistency.

The science: Thickeners like cornstarch can create dense layers that reduce heat penetration during canning, resulting in under-processed and unsafe preserves. Canning recipes have been tested without added thickeners to ensure proper heat processing.

Source: PNW 395 Salsa recipes for canning

06-Jams-005 **FALSE** **HIGH RISK**

Myth: You should never reduce the sugar in a jam recipe because it will always make the product unsafe.

Bust: You can reduce sugar safely in jam recipes, but you must use special low- or no-sugar pectins and follow adapted recipes carefully. Simply cutting sugar in a traditional recipe can cause jams to have poor texture and possibly be less safe. Using tested products and instructions for low-sugar jams ensures both good quality and safety.

Why believed: Traditional jam recipes have always used a lot of sugar, which acts as a preservative. This has led many to think sugar reduction is unsafe without considering modern pectin technology.

The science: Sugar is traditionally important for gelling and preservation in jams. However, specialized low-methoxyl pectins form gels without high sugar and with added calcium, maintaining safety if proper procedures are followed. These products are tested to ensure adequate acidity and gel formation to prevent bacterial growth.

Source: SP 50-632 Making berry syrups at home

FREEZING & COLD STORAGE

06-Freezing-001 **TRUE** **LOW RISK**

Myth: Properly freezing seafood immediately after catching preserves its best texture and flavor, even if rigor mortis isn't complete.

Bust: Freezing fish soon after catching and icing it slowly through rigor mortis maintains quality better than freezing fish that has warmed or gone quickly through rigor. Poor handling before freezing causes toughness or off-flavors that freezing can't fix.

Why believed: Some think fish must fully 'relax' after rigor mortis before freezing to avoid damage.

The science: Fish frozen before rigor mortis passes slowly through rigor while frozen, resulting in better flavor and texture. Rapid cooling after catch by icing delays harsh rigor and reduces muscle damage.

Source: PNW 586 Home freezing seafood

06-Freezing-002 **TRUE** **LOW RISK**

Myth: Storing fruits and vegetables at refrigerator temperatures below 40°F slows spoilage and maintains nutrition for many months in the freezer.

Bust: Most fruits and vegetables remain high quality for 8 to 12 months when frozen at 0°F (-18°C) or below. Freezing at proper temperatures preserves nutrition and slows quality loss.

Why believed: People sometimes doubt if freezing preserves flavor and nutrients well or worry that freezing damages produce over time.

The science: Freezing freezes water to stop microbial growth and enzyme activity, slowing spoilage and nutrient loss. Storing at consistent 0°F or lower using a thermometer ensures best long-term quality.

Source: PNW 214 Freezing fruits and vegetables

06-Freezing-003 **FALSE** **HIGH RISK**

Myth: Once frozen, you can safely refreeze any thawed food as long as it looks and smells fine.

Bust: Refreezing thawed food is not always safe. If the food has warmed above 40°F (5°C) during thawing, bacteria may have grown to unsafe levels, even if it looks or smells normal. Refreezing can trap bacteria and increase the risk of foodborne illness.

Why believed: Many people assume freezing kills all bacteria and that freezing again doesn't affect safety if the food seems okay.

The science: Freezing stops the growth of bacteria but does not kill all bacteria present. Once thawed, if food stays above 40°F for too long, bacteria can multiply rapidly. Refreezing does not eliminate these bacteria, potentially causing illness. Safe

refreezing is only for foods that still contain ice crystals and have been kept at safe temperatures.
Source: PNW 214 Freezing fruits and vegetables

06-Freezing-004 **FALSE** **HIGH RISK**

Myth: You should always discard canned foods or jars that have frozen, as freezing can make them unsafe.

Bust: Frozen canned foods can still be safe if the container remains intact. Freezing may affect texture or cause bulging, but safety depends on the integrity of the can or jar. Carefully check for seals and damage before deciding to discard.

Why believed: Freezing can cause cans or jars to bulge or break, so it's commonly thought all frozen cans are unsafe.

The science: If the seams are intact and the food is still cold, the food can be safely salvaged by refrigeration or refreezing. Broken seals or bulging combined with thawing to room temperature are signs to discard. Boiling low-acid foods after freezing adds extra safety.

Source: PNW 296 Freezing convenience foods that you've prepared at home

06-Freezing-005 **FALSE** **HIGH RISK**

Myth: Cooling large batches of hot food in the refrigerator is unsafe because it raises the fridge temperature and risks spoilage.

Bust: You can safely refrigerate large amounts of hot food if you speed cooling by dividing it into shallow containers or chilling in ice water first. This prevents the food from staying in the danger zone too long and protects all groceries in the fridge.

Why believed: People worry that putting hot food directly into the refrigerator warms the fridge and other foods, causing spoilage.

The science: Promptly cooling food prevents bacterial growth. Using shallow containers and ice baths helps bring the temperature down quickly to below 40°F. Proper airflow in the refrigerator is important but avoiding prompt cooling risks foodborne illness.

Source: PNW 296 Freezing convenience foods that you've prepared at home

DEHYDRATING & SMOKING

06-Drying-001 **TRUE** **HIGH RISK**

Myth: Pre-cooking or heating meat to 160°F before or immediately after dehydrating jerky is necessary to make jerky safe for storage and consumption.

Bust: Cooking meat to 160°F kills harmful pathogens like Salmonella and E. coli that might be present. Drying alone at low temperatures may not destroy these bacteria, so heating before or after dehydration ensures safer jerky.

Why believed: Some think drying at low temperatures is all that's needed because the meat becomes dry and leathery, and visible dryness indicates safety. This overlooks microbial survival.

The science: Studies show pathogens can survive dehydration alone. A heat step reaching 160°F before or after drying kills these microorganisms, making the jerky shelf-stable and safely edible.

Source: PNW 632 Making jerky at home safely

06-Drying-002 **TRUE** **LOW RISK**

Myth: Allowing fish to air dry until a sticky pellicle forms before smoking helps smoke to adhere and improves safety.

Bust: Forming a pellicle by air drying fish before smoking is an important step. It helps smoke flavors adhere evenly and also reduces the risk of surface spoilage during the smoking process.

Why believed: This step is a traditional smoking practice passed down in many recipes to improve texture and flavor, often described as essential for good smoked fish.

The science: A pellicle forms a slightly tacky surface on fish that helps trap smoke compounds, enhancing flavor and preservation. Adequate drying also discourages bacterial growth at the surface.

Source: PNW 238 Smoking fish at home

06-Drying-003 **FALSE** **HIGH RISK**

Myth: Dehydrating jerky at any low temperature is safe as long as the meat feels dry and leathery at the end.

Bust: Jerky must be dried at the right temperature (usually at least 130°F) to ensure harmful bacteria are destroyed. Just feeling dry is not enough to guarantee safety. If dried too slowly or at too low a temperature, bacteria may survive and multiply, leading to unsafe food.

Why believed: People often base safety on texture and dryness because dried appearance seems like preservation, and old family jerky recipes often lack temperature guidance.

The science: Meat jerky must be heated to at least 130°F and sometimes preheated before drying to kill bacteria such as Salmonella and E. coli. Simply drying until leathery without these controls risks foodborne illness.
Source: PNW 632 Making jerky at home safely

06-Drying-004 **FALSE** **HIGH RISK**

Myth: Smoking fish at low temperatures below 90°F can preserve fish safely for long periods without refrigeration.

Bust: Cold smoking below 90°F does not cook or kill bacteria effectively, so fish smoked this way must be refrigerated or frozen. Relying on smoke alone at low temperatures is unsafe as it does not prevent bacterial growth that causes spoilage and illness.

Why believed: Cold smoking is a traditional method thought to add preservative smoke flavor, and many recipes mention 'cold smoking' for flavor without temperature warnings.

The science: Smoke itself is not a reliable preservative; low temperatures do not kill bacteria. Hot smoking must reach at least 150°F internal temperature to ensure safety. Without adequate heat, bacteria grow, spoiling fish and risking food poisoning.
Source: PNW 238 Smoking fish at home

06-Drying-005 **FALSE** **HIGH RISK**

Myth: Smoking alone preserves meat and fish safely without need for refrigeration or cooking afterward.

Bust: Smoking adds flavor and some preservation qualities but is not sufficient to make meat or fish shelf-stable. Hot smoking should include cooking the product to a safe internal temperature, and smoked products usually require refrigeration or freezing.

Why believed: Many traditional recipes suggest smoking preserves food for long storage, and commercial smoked meats are often refrigerated, a detail home cooks miss.

The science: Smoke contains antimicrobial compounds but alone doesn't eliminate pathogens. Hot smoking requires at least 150°F internal temperature for 30 minutes to cook and kill bacteria. Without proper cooking and refrigeration, smoked foods spoil.
Source: PNW 238 Smoking fish at home